

Trying to see whether a one-assignment repair actually helps greedy, and how much slower it is. Start with one frozen instant. Each target is worth one; satellites have beam limits. Links need enough elevation, and beams from the same satellite need enough angular separation.

For greedy: go through targets in input order, try the closest satellite that still works. Use satellite index to break a distance tie.

Example to reproduce in code. Set Earth's radius to 1. This puts the satellites unrealistically high, but makes the arithmetic easy. Everything is on the equator:

	A	B	S_1	S_2
longitude	0°	30°	15°	-30°
radius	1	1	2	2

Capacity 1 per satellite; minimum elevation 20° . For longitude difference Δ ,

$$\text{distance} = \sqrt{5 - 4 \cos \Delta}, \quad \sin e = \frac{2 \cos \Delta - 1}{\sqrt{5 - 4 \cos \Delta}}.$$

The useful numbers:

	distance	elevation
$A \rightarrow S_1$	1.066	60.95°
$A \rightarrow S_2$	1.239	36.21°
$B \rightarrow S_1$	1.066	60.95°
$B \rightarrow S_2$	1.732	0°

Process A first: it takes S_1 because that is closer. B can only see S_1 , so greedy serves 1. Move A to S_2 and give S_1 to B : now it serves 2. This example doesn't exercise beam separation since both satellites have capacity 1.